

Nex HMIS: General Overview for All Stakeholders

What is the Nex HMIS?

The Kenya Hospital Management Information System (HMIS) is a user-friendly, secure digital platform designed to transform healthcare delivery in Kenya's hospitals and clinics, both urban and rural. It streamlines patient care, administrative tasks, and compliance with national standards, working even in areas with limited internet. The system integrates with Kenya's health platforms like KHIS (DHIS2) for reporting and the Social Health Authority (SHA) for insurance claims, ensuring compliance with the Kenya Data Protection Act (2019).

Value for Stakeholders

- **Patients:** Get faster, coordinated care with secure data privacy.
- **Clinicians:** Focus on care with easy access to patient records and lab results.
- **Nurses:** Log vitals and encounters quickly, even offline, improving workflow.
- **Front Desk Staff:** Register patients and manage records efficiently, reducing wait times.
- **Pharmacists:** Track medications and avoid stockouts with real-time alerts.
- **Cashiers/Claims Officers:** Process payments and SHA claims seamlessly.
- **Administrators:** Monitor hospital performance and ensure regulatory compliance.
- **IT Teams:** Maintain a reliable, scalable system with minimal effort.
- **Regulators:** Receive accurate, timely data for national health oversight.

Key Capabilities

1. **Patient Management:** Register patients using national ID, phone, or unique Medical Record Number (MRN). Search records quickly and track consent for data sharing.
 - *For Front Desk:* Register patients in seconds, reducing queues.
 - *Example:* A clerk enters a patient's phone number, and the system generates an MRN for tracking.
2. **Clinical Tools:** Record vitals, diagnoses, and lab/radiology orders with real-time results.
 - *For Clinicians/Nurses:* Log patient data and view test results instantly.
 - *Example:* A nurse records blood pressure offline during outreach, and a clinician reviews lab results on the web app.
3. **Pharmacy & Inventory:** Manage medication stock, prescriptions, and dispensing with alerts for low stock or expirations.
 - *For Pharmacists:* Update stock and dispense medications accurately.
 - *Example:* A pharmacist checks stock, dispenses an antibiotic, and gets an alert for low inventory.
4. **Billing & Claims:** Generate invoices, process payments (e.g., M-Pesa), and submit SHA claims with eligibility checks.
 - *For Cashiers/Claims Officers:* Handle payments and claims without delays.

- *Example:* A cashier records an M-Pesa payment, and a claims officer submits an SHA claim with lab reports.
- 5. **Reporting:** Automate KHIS reports (e.g., OPD visits, immunizations) and create dashboards for hospital insights.
 - *For Administrators:* Track performance and meet reporting requirements effortlessly.
 - *Example:* OPD data is compiled into a KHIS report automatically, saving hours of work.
- 6. **Offline Access:** Use the mobile app to log data in remote areas, syncing when connected.
 - *For Nurses/Clinicians:* Work seamlessly in rural clinics without internet.
 - *Example:* A nurse logs vitals offline, and the data syncs when back at the facility.
- 7. **Interoperability:** Share data with other systems via FHIR for care coordination and national integration.
 - *For IT/Claims Officers:* Ensure smooth data exchange with KHIS and SHA.
 - *Example:* A patient's record is shared with another hospital, ensuring continuity of care.
- 8. **Security & Privacy:** Protect data with role-based access, encryption, and audit logs, meeting Kenya's privacy laws.
 - *For All:* Trust that patient data is secure and access is controlled.
 - *Example:* Only authorized clinicians view sensitive HIV records, with all access logged.
- 9. **Audit & Monitoring:** Track user actions and system performance for transparency and troubleshooting.
 - *For Administrators/IT:* Ensure accountability and system reliability.
 - *Example:* An audit log identifies who accessed a patient's record, resolving a privacy query.
- 10. **Future AI Tools:** Predict no-shows, stockouts, claim denials, and clinical risks (e.g., sepsis).
 - *For All:* Improve planning and patient care with smart insights.
 - *Example:* The system flags a patient for sepsis risk, prompting a clinician's review.

Scope & Timeline

- **Phase 0 (Now):** Setting up infrastructure and security.
- **Phase 1:** Core features (patient records, clinical tools, pharmacy, billing).
- **Phase 2:** SHA claims, theatre scheduling, enhanced reporting.
- **Phase 3:** Maternal/child health, immunizations, analytics.
- **Phase 4:** AI-driven predictions and alerts.

The system scales from single clinics to regional networks, built with modern tech (Django, Next.js, React Native, PostgreSQL) for reliability.

Why It Matters

The Nex HMIS simplifies tasks, reduces errors, and improves patient care. It saves time for clinicians, nurses, and staff, ensures compliance for administrators, and supports Kenya's health goals with accurate data. From rural clinics to urban hospitals, it's a tool that empowers everyone to deliver better healthcare.

Key Features for Clinicians

1. **Patient Management (PAS - Patient Administration System)**
 - Easily register and find patients using identifiers like national ID, phone number, or a unique medical record number (MRN).
 - View patient details, including demographics, consent for data sharing, and flags for sensitive cases (e.g., HIV or GBV), with restricted access to protect privacy.
 - Example: You can quickly pull up a patient's history before a consultation, ensuring you have their full medical context.
2. **Clinical Encounters and Observations**
 - Record patient encounters, including vitals (e.g., blood pressure, temperature), diagnoses, and procedures, all in one place.
 - Supports standardized coding (e.g., LOINC for vitals) for consistency and interoperability with other systems.
 - Example: During a ward round, you can log a patient's vitals or diagnoses directly on the mobile app, even if there's no internet, and it will sync later.
3. **Orders and Results**
 - Request lab tests or radiology directly from the system, track their status, and view results when ready.
 - Example: Order a blood test for a patient, and get notified when the results are available, reducing delays in decision-making.
4. **Pharmacy and Medication**
 - Check medication stock levels, request prescriptions, and track dispensing.
 - Alerts for low stock or expiring medications to avoid disruptions.
 - Example: Prescribe medication and confirm it's available in the pharmacy without leaving the patient's record.
5. **Billing and Claims Integration**
 - Generate invoices for services and link them to SHA claims for insured patients, reducing administrative burden.
 - Example: After a consultation, the system automatically prepares billing details, and clerks can handle SHA claim submissions.
6. **Offline Capability**
 - The mobile app works offline, allowing you to record vitals, notes, or orders in rural areas with poor connectivity. Data syncs automatically when a connection is available.
 - Example: In a remote clinic, you can document a patient's visit and sync it later without losing information.
7. **Reporting for KHIS**
 - Automatically compiles data (e.g., OPD visits, immunizations) for mandatory KHIS reporting, saving you time on paperwork.
 - Example: Your patient encounter data feeds into national health indicators without manual reporting.
8. **Privacy and Security**
 - Built with strict access controls to protect sensitive patient data (e.g., only authorized clinicians can view HIV-related records).
 - Every action is logged for accountability, aligning with Kenya's Data Protection Act.
 - Example: You can trust that patient data is secure and only accessible to those with proper permissions.

How It Supports Your Workflow

- **Efficiency:** Quickly access patient records, order tests, or prescribe medications from a single interface, reducing time spent on paperwork.
- **Accuracy:** Standardized data entry minimizes errors, and real-time access to lab results or medication stock improves decision-making.
- **Mobility:** Use the mobile app during ward rounds or community outreach, even offline, to keep records up to date.
- **Collaboration:** Share patient data securely with other clinicians or departments (e.g., pharmacy, lab) within the system.
- **Compliance:** The system handles regulatory requirements (e.g., KHIS, SHA, data protection) so you can focus on patient care.

Example Scenario

Imagine you're seeing a patient in a rural clinic:

1. You search for the patient by their phone number or national ID on the mobile app.
2. You review their medical history, note a new encounter, and record vitals like blood pressure.
3. You order a lab test and prescribe medication, checking stock availability instantly.
4. If offline, the app saves your entries locally and syncs them when you're back online.
5. The encounter data automatically feeds into KHIS reports, and billing is prepared for SHA claims without extra effort.

Future Enhancements

In later phases, we plan to add:

- **AI-Powered Insights:** Alerts for risks like sepsis based on vitals trends or predictions for appointment no-shows to optimize scheduling.
- **MCH and Immunization Tracking:** Specialized modules for maternal and child health programs.
- **Advanced Analytics:** Dashboards to track your clinic's performance, like patient load or common diagnoses.

Your Input Needed

As a clinician, your perspective is critical to making this system practical and effective. We'd love your feedback on:

- **Usability:** Are the proposed features (e.g., encounter logging, offline mode) intuitive for your workflow?
- **Clinical Needs:** Are there specific tools or data points (e.g., specific vital signs, diagnostic codes) you need?
- **Challenges:** What pain points in your current workflow should the system address (e.g., delays in lab results, manual reporting)?
- **Mobile Use:** How do you envision using the mobile app in your daily tasks?

- **Training:** What kind of support or training would help you and your team adopt this system?